Physical AI Oncology Trial Protocol

Sponsor: ChemicalQDevice (illustrative)
Phase: feasibility
Population: adults with solid tumors
Intervention: robot-assisted procedure under a verification gate
Primary endpoint: procedure-level safety with the gate cleared before execution

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Abstract.

This protocol template describes a Physical AI oncology trial in which an automated verification gate clears robot-patient interaction code before it is executed. It is a template: each section carries one short paragraph to be expanded.

Disclaimer: This work is independent and not endorsed or sponsored by trial sponsors, FDA, CROs, sites, IRBs, regulators, or medical societies; and was generated using Artificial Intelligence.

Keywords

1 Background and rationale

Robot-assisted oncology procedures increasingly rely on machine-generated control code, and the protocol's premise is that verification must precede generation. The rationale rests on demonstrated feasibility of an automated gate.

2 Objectives and endpoints

The trial has one primary and two secondary objectives.

1. Primary: establish procedure-level safety with the verification gate cleared before execution.
2. Secondary: characterize the gate's escalation rate to a qualified human.
3. Secondary: quantify run-to-run variation against the coefficient-of-variation bound.

3 Eligibility

Eligibility, consent, and withdrawal follow 21 CFR part 50 and the harmonized good clinical practice framework; the verification gate does not alter human-subject protections.

Figure placeholder (`Images/trial-schema.png`). The float, caption, and `\label` mirror the VVUQ-02 figures; replace the box with `\includegraphics`.

Figure 1: Figure 1. Placeholder trial schema from screening through gated intervention to follow-up.

4 Endpoints and assessments

Endpoint	**Type**	**Timing**
gate cleared before execution	primary	each procedure
escalation to human	secondary	each procedure
run-to-run variation	secondary	per sweep

Table 1: Table 1. Endpoints and assessment timing (modeled on VVUQ-05 Table 1).

5 Statistical considerations

Analysis is descriptive for this feasibility template; the verification fraction is reported as an exact proportion and must equal 1.0 for an interaction to proceed.

Acknowledgments

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Ethical Disclosures

The author declares no competing interests. This is an independent draft and is not endorsed, sponsored, or approved by any trial sponsor, CRO, site, IRB, regulator, or medical society, nor by CFR, ICH, or FDA. All supporting numbers are simulation or illustrative results.

Cite This Protocol

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Data Availability

Supporting records are openly available in the `kevinkawchak/cancer-automated` and `kevinkawchak/physical-ai-oncology-trials` GitHub repositories and are archived on Zenodo. All data sets are synthetic or illustrative and reproducible from a deterministic seed; no patient-identifiable information is included.

References

[1] K. Kawchak, "VVUQ Processing Priority over Code Generation: Method and Pipeline." `https://doi.org/10.5281/zenodo.20372501`, 2026. `https://doi.org/10.5281/zenodo.20372501`.

[2] K. Kawchak, "Mobile Pancreatic Cancer Unitree H2 Surgical Humanoid with Priority VVUQ." `https://doi.org/10.5281/zenodo.20421754`, 2026. `https://doi.org/10.5281/zenodo.20421754`.

[3] K. Kawchak, "Verification Before Generation in Physical AI Oncology Trials Act of 2026 (H. R. 9510)." `https://doi.org/10.5281/zenodo.20535429`, 2026. `https://doi.org/10.5281/zenodo.20535429`.

[4] K. Kawchak, "National Physical AI Oncology Trial Platform." `https://doi.org/10.5281/zenodo.19244918`, 2026. `https://doi.org/10.5281/zenodo.19244918`.

[5] Creative Commons, "Attribution 4.0 International (CC BY 4.0)." `https://creativecommons.org/licenses/by/4.0/`, 2013. `https://creativecommons.org/licenses/by/4.0/`.

[6] L. Lamport, *LaTeX: A Document Preparation System*. Addison-Wesley, 2nd ed., 1994.